



DIGITAL DEVICES

Pearson Edexcel
ICT 9-1

by teacher
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COMPUTERS



MOBILE
DEVICES



PERIPHERAL
DEVICES



INPUT & OUTPUT
DEVICES



DIGITAL
TECHNOLOGY
EVERYWHERE

Introduction

Digital devices are pieces of hardware that use computers or microcontrollers, and they are found everywhere in our digital world. They enhance and support how we live our lives every day. They can connect and work together to give us the data we need, when and where we need it. Digital devices are always developing. This changes the way in which they are used by individuals, organisations and local, national and global societies.

TYPES OF DIGITAL DEVICES

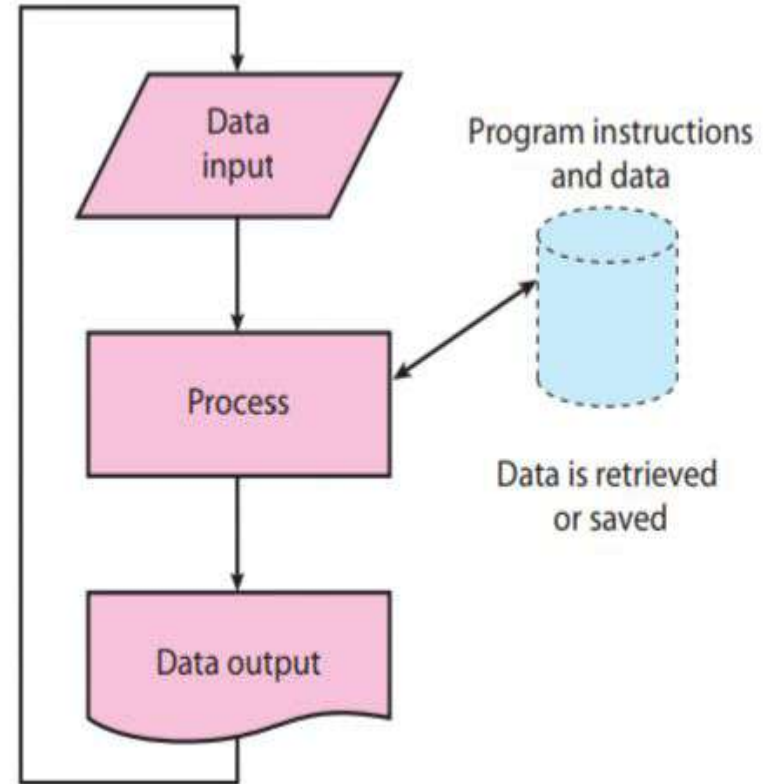
- Computers
- Mobile phones
- Tablets
- Cameras and camcorders
- Home entertainment systems
- Navigation aids
- Home automation devices and smart assistants

What is a computer?

- A computer is an electronic machine that **inputs data**, **processes** it under the control of a stored program, and **outputs information**.
- While the data is being processed, data can be retrieved from or saved on backing **storage**
- **Data** is raw, unprocessed information. Information is data that people understand. In order to understand data, you may have to interpret it:
- The number 30112012 is data that can be interpreted as:
 - A date 30/11/2012
 - A sum of money \$301,120.12
- The interpreted data, that is the date or the sum of money, is the information.

- **Input** – data is entered into the computer.
- **Process** (turn data into information) - computer is controlled by a program, that is, a sequence of instructions. It processes the input data automatically following these instructions.
- **Store** – the program and data are stored, e.g. on a hard disk, for later retrieval.
- **Output** – the computer communicates to the user, e.g. it displays graphics on the screen.

Computer model



Types of computer

Computers can be:

- **PCs (personal computers)** – a microcomputer for individual use.
- **Mainframe computers** – large computer, with huge processing power.
- **Minicomputers** – smaller version of a mainframe.
- **Supercomputers** – very large mainframe.



N/B: Mainframe is a large, powerful computer that can do a lot of complicated jobs quickly and can be used by a lot of people at the same time.

Different types of PC

- **Desktop** – has these basic components: monitor, keyboard, system unit and mouse
- Some desktops are “all-in-one” meaning they combine the monitor with pc hardware.



- **Laptop** – a portable computer slightly larger than A4 size.
- Some laptops are desktop replacements (they are bigger than other laptops and have a better screen resolution and performance than other laptops)



Embedded computers

An embedded computer is designed for and built into a specific application where it will perform a limited range of dedicated functions. The size and functionality of an embedded computer depends on the application. They may be very small devices built into a single microchip and may control the device.



Hand-held computer

A handheld computer or Personal Digital Assistant (PDA) or palmtop can fit in one hand or in your pocket, but it is too small for general work. A PDA usually has a touch sensitive screen. Although PDAs can be temporarily attached to a Keyboard, you cannot comfortably type a long document into a PDA.



Mobile phones

- Mobile phones use SIM (subscribers identity module) to connect to mobile phone networks
- **Specialists phone** have special features to provide users with functions that meet particular user needs



Smartphone

- Smartphones are small computers with Wi-Fi and mobile phone connectivity to allow them to make phone calls and access the internet.
- They also include features of other devices such as cameras, media players and hand-held games consoles.
- They have a more advanced operating system than other mobile phones.
- Applications (or apps) can be downloaded onto the smartphone, which allow users to customise their smartphones with entertainment, educational and business features.
- Most smartphones use a touch screen to allow users to input information. A virtual keyboard is used to enter text, numbers and other characters.
- Because they combine so many features, smartphones use more power than other types of mobile phone. This means that they have a shorter battery life and need to be charged more regularly than other mobile phones.

Tablets

- Tablet devices or tablets are bigger than smartphones, but have similar features. For example, a tablet device has a touch screen, apps and Wi-Fi Connectivity to provide access to the internet.
- Some table devices have SIM card slots to allow internet connectivity using the moble phone network, so that they can be online when they are not within range of a Wi-Fi signal.



CAMERAS AND CAMCORDERS

- Digital cameras and camcorders use light sensors to capture images formed by light passing through the device's lens.
- Traditionally, cameras are used to capture still images and camcorders are used to capture moving images.
- However, most digital cameras can now film moving images and most camcorders can photograph still images.



Home entertainment systems

They include :

- ❑ Televisions
- ❑ Sound systems
- ❑ Personal video recorder
- ❑ Blue-ray and DVD player
- ❑ Game consoles
- ❑ Media players



Televisions



- Televisions display still and moving images on a screen.
- The quality of the image is set by the number of pixels that are used to display the image. This is referred to as the screen's resolution.
- resolution is the number of pixels used by a screen to display an image
- Pixels are small dots that helps to make up an image

SOUND SYSTEMS

Sound systems can produce loud, rich sound using high-quality speakers and amplifier. Some speakers contain built-in amplifiers



MEDIA PLAYERS

They are electronic devices that can store and play digital music and videos and show digital photographs



PERSONAL VIDEO RECORDERS



A personal video recorder (PVR) is a device that records broadcasted content so that it can be watched at a later date.

BLU-RAY AND DVD PLAYERS

Blu-ray and DVD players connect to televisions in order to play films and other content that is stored on DVD or Blu-ray disks.



▲ Figure 1.15 A Blu-ray player

GAMES CONSOLES



Games consoles are designed to enable users to play video games on a television screen. Games are provided on disks or as downloads from the internet. They use controllers, which are often wireless, to control the characters, vehicles and/or objects in the game.

Some consoles use motion sensors to allow the player to control the game with gestures and body movements.

Other games use virtual reality controllers and headsets to immerse the player in a realistic gaming experience, where their own movements in the real world are replicated by an avatar in the virtual world of the game.

NAVIGATION AIDS

Navigation aids (such as Sat-Nav) can calculate the best route between two or more locations and can provide updates to the route if it is not followed accurately. They are commonly used in caravans , delivery vans and ships.

They provide visual prompt and alerts to help drivers take the correct route, such as by taking a particular turning. Specialist devices with audio alerts and waterproof and shockproof cases are available for walkers, cyclists and runners.



- ❑ GPS is the Global Positioning System that uses radio signals from satellites to show your exact position on the Earth on a special piece of equipment
- ❑ Navigation aids use information from GPS satellites to determine the exact location of the device on Earth.
- ❑ Navigation aids also use orientation sensors to know which way the device is currently pointing, and the device will display the user's position and orientation on a map. Maps are either stored permanently on the device or downloaded from the internet when they are needed.
- ❑ GPS does not need Internet connectivity in order to work. However, navigation aids need internet connectivity in order to update map data.

HOME AUTOMATION DEVICES AND SMART ASSISTANTS



One type of emerging technology is a group of devices that can be used to create smart homes. For example home automation devices can connect a range of digital devices which sense and control functions in the home, such as temperature and lighting. These functions can be controlled from apps on smartphones or the internet.

FEATURES OF DIGITAL DEVICES

- They include:
 - Portability
 - Performance
 - User interface
 - Security features
 - Energy consumption
 - Expansion capabilities
 - Connectivity
 - Media support
 - Storage

PORTABILITY

- Portability is the ability of a device to be easily carried or moved around.
- For a device to be portable, it needs to be easy to carry and move around.
- This means that the portability of a device is directly related to its size and weight.
- For some devices, such as a television or a desktop computer, portability is unlikely to be a priority.

STORAGE

- Files and programs are stored in storage.
- More available storage allows users to store more files and programs.
- Storage speed also affects performance

CONNECTIVITY

- Devices can share data by connecting to each other using wired or wireless connectivity.
- Connectivity can be used to update software. back up files or play media from one device on another.
- Different connectivity types provide different speeds of data transfer and levels of convenience.
- Wired connectivity is usually faster and more reliable, but introduces additional cost, mess, inconvenience and safety risks such as tripping especially for young children.

MEDIA SUPPORT

- Different devices can read data from and write data to different types of media.
- Examples of media include SD and microSD flash memory cards, and DVD.
- If devices do not have built-in (native) media, adapters can usually be connected to provide connectivity to an external device into which the media can be inserted.



EXPANSION CAPABILITY

- Some PCs allow users to install additional components.
- Some smartphones and tablet devices have expansion slots to allow them to make use of flash memory cards.
- Systems can also be expanded using ports such as USB ports. These allow the user to connect extra devices called peripheral

ENERGY CONSUMPTION

- Digital devices require electricity to work. One benefit of lower energy consumption in mobile devices is a longer battery life.
- Due to the rising cost of energy and pressure from customers and governments to be more environmentally responsible, manufacturers are creating digital devices that consume less energy.
- The aim of this is to save their customers money and demonstrating social responsibility, such as by reducing environmental damage

PERFORMANCE

- A high-performance device performs its job or tasks quickly.
- The speed at which a device performs is determined by the speed at which it can carry out instructions from its software.
- Software instructions are carried out by the processor, which means that a faster processor will increase the performance of the device.

- RAM (Random Access Memory), which is the memory in a computer system that is used for running software. Instructions are loaded into the processor from RAM, which means that faster RAM will also allow instructions to be loaded into the processor more quickly.
- RAM holds all software instructions to be carried out, so more RAM will allow the system to have more programs running at the same time.

SECURITY FEATURES

- The data stored on digital devices may be private, valuable or both.
- This means that devices need to have security features to keep their users' data safe.

SOFTWARE SECURITY

- They prevent unauthorised access to data.
- digital devices have a range of software security features to ensure that the person trying to use the data is authorised.
- Examples of software security features are use of:
 - passwords,
 - PIN (personal identification numbers) and
 - biometric systems.



PHYSICAL SECURITY

- Physical security to prevent theft is also important.
- Many devices use security slots which can have locks attached to secure them to furniture.
- Some attach to specialist slots. Others connect to a port on the device with special screws used to secure the locks in place.



USER INTERFACE

- User interface is the device's ability to interact with a user
- Users give commands to a device through the user interface. There are several different types of user interface including :
 - ❑ Command line
 - ❑ Graphical user interface
 - ❑ Menu-driven
 - ❑ Voice interface
 - ❑ Gesture interface

COMMAND LINE INTERFACE (CLI)

When using a command line interface, users enter text instructions and the computer system provides results or feedback as text.



```
Command Prompt
C:\>dir /?
Displays a list of files and subdirectories in a directory.

DIR [drive:][path][filename] [/A[:attributes]] [/B] [/C] [/D] [/O[:sortorder]] [/P] [/Q] [/S] [/T[:timefield]] [/W] [/X] [

[drive:][path][filename]
    Specifies drive, directory, and/or files to list.

/A      Displays files with specified attributes.
attributes  D Directories          R Read-only files
              H Hidden files        A Files ready for ar
              S System files        - Prefix meaning not

/B      Uses bare format (no heading information or summary)
/C      Display the thousand separator in file sizes. This
              default. Use /-C to disable display of separator.
/D      Same as /B but files are list sorted by column.
/L      Uses lowercase.
/N      New long list format where filenames are on the far
/O      List by files in sorted order.
sortorder  N By name (alphabetic)    S By size (smallest
              E By extension (alphabetic) D By date/time (olde
              G Group directories first - Prefix to reverse

/P      Pauses after each screenful of information.
/Q      Display the owner of the file.
/S      Displays files in specified directory and all subdir
/T      Controls which time field displayed or used for sort
timefield  C Creation
              A Last Access
              W Last Written
/W      Uses wide list format.

Press any key to continue . . .
```

MENU-DRIVEN INTERFACE

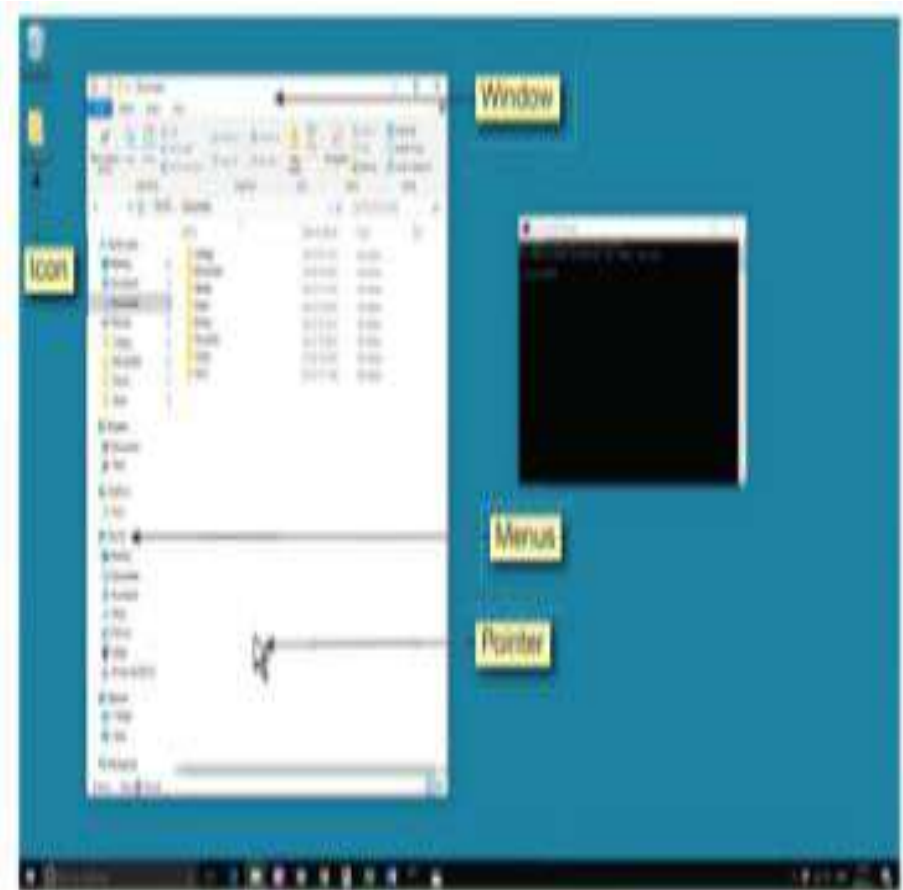
- This type of interface displays a list of options as a menu.
- Selecting one of these options will either trigger a command or display another menu with further options to choose from.
- Menu-driven interfaces are easier to use than command line interfaces, but can take more time as you have to go through the menu structure each time you want to carry out a command, rather than typing the command directly into the system.
- This type of interface is used in many devices including ATMS (cash machines in banking), televisions and older mobile phones.



GRAPHICAL USER INTERFACE (GUI)

- A graphical user interface is controlled by a pointer on the screen and uses a screen made up of windows, icons and menus.
- **Windows** are areas of the screen that are dedicated to applications or operating system tasks
- **Icons** are small images that represent an application. They can be selected with the pointer to open the application.

- **Menus** provide options for tasks relating to the operating system or open applications.
- This is the easiest type of interface to use, but it takes up more memory and storage than a command line or menu-driven interface.



VOICE INTERFACE

- A voice interface allows the user to give spoken commands to a device.
- The device has voice recognition software which matches the spoken words against a library of words to find a match. To save storage on the device, the library of words is often stored online, so these devices usually require internet access.
- The disadvantage of using a voice interface is that sometimes the software cannot find a match or returns an incorrect match, which produces unwanted results

GESTURE INTERFACE

- This type of interface allows the user to control the device by swiping their finger or fingers across the screen, or by pinching their fingers together to 20cm in or out.
- This type of interface is commonly found on devices with touch screens.

TYPES OF PERIPHERAL DEVICES

- Peripheral devices are devices that can be connected to a computer, such as a PC or tablet device.
- Peripherals can be connected inside or outside a computer and can be grouped into three types:
 - input
 - Output
 - storage.



Input

- Input peripherals are devices that send data to the computer.
- They allow the user to control the computer or store data captured from sources outside the system.



Keyboard



Graphic pad



Web cam



Mouse



Scanner



Joystick



Track Ball



Digital Camera



Microphone

KEYBOARDS

- Keyboards use buttons known as keys, which users press to input text or to interact with software.
- Keyboards send data to the computer either using a wired or wireless connection.
- Different types of keyboards are used for different operating systems, languages and functions.



POINTING DEVICES

A **pointer** is used on a screen to select displayed objects. There are several types of device that allow you to control a pointer.

Mouse:

- A mouse uses an optical sensor to recognise the movement of the device.
- Its sensitivity can be affected by the surface upon which it is placed
- Some mice use roller balls to control the movement of the pointer.



Tracker ball:

- A tracker ball is rolled to move the pointer.
- It does not have buttons to press.



Track pad:

- The track pad surface senses finger movements, touches and presses.
- Track pads are often built into laptop computers to save space





Joysticks:

- Joysticks are commonly used on games controllers to move around the sprites.
- Sprites are characters in a game

Graphics tablet:

- A graphics tablet is a flat pad that is used with a stylus.
- These devices are often used by digital artists and designers.



SCANNER (INCLUDING OCR AND OMR)

- Scanners use light sensors to record physical documents as images, which are saved as files in a computer.
- Software allows scanners to read characters on a document and store the results in a text file. This is known as **Optical Character Recognition (OCR)**
- **Optical Mark Recognition (OMR)** software can also be used with scanners to detect simple marks in a document.
- A common use for OMR is for recognising and recording responses to multiple choice tests

[illegible]

OMR detects simple marks on a document

[illegible]

BARCODE SCANNER

- A barcode is a pattern of lines and gaps that can be read by a barcode scanner, which detects the width of lines and gaps in a barcode .
- Barcode are often used on parcels, so that they can be tracked, and on items for sale in shops.
- The barcode represents letters and numbers which are used to identify the item.
- There are two types of barcodes: linear and matrix.
- Matrix barcodes are also known as QR codes. They are newer than linear

barcodes and have some advantages over linear barcodes, such

- they can hold much more information than linear barcodes
- they can be scanned from any angle.



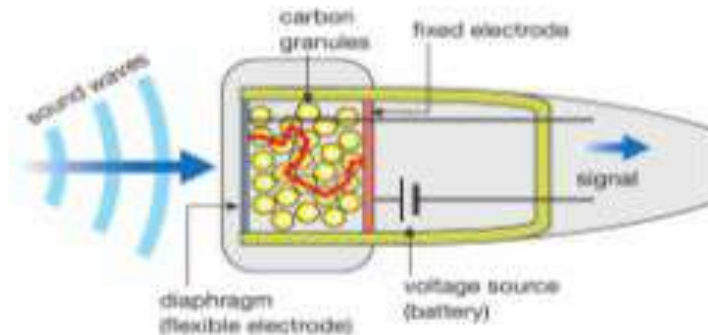
WEBCAM

- Webcams are specialised cameras.
- They are generally lower quality than camcorders, and may have built-in microphones to capture sound.
- They can be used as security cameras and can stream images or video to the internet.



MICROPHONE

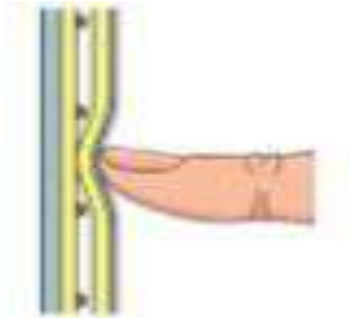
- Microphones are used to capture sound.
- It uses a diaphragm which moves when air hits it. This movement produces an analog electric signal.
- A microphone converts the electrical signal to a digital signal with its sound Card, which acts as an analog-to-digital converter



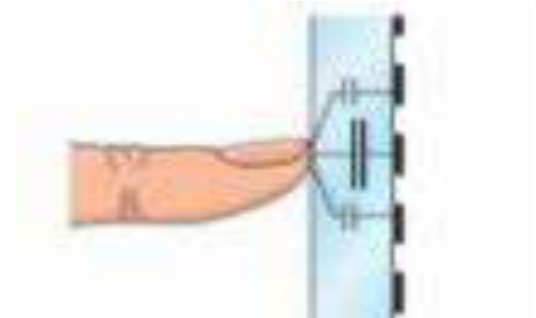
TOUCH SCREEN

- Touch screens are used in many devices, including smartphones, tablet devices, laptops and desktop Computers.
- There are two types two touch screen:

(1) Resistive



(2) Capacitive



Resistive.

- When a user presses on a resistive screen, the pressure causes two layers underneath the screen to touch and make a connection.
- Resistive touch screens are more durable than capacitive touch screens, but are harder to read because more layers reflect more light.
- In addition, they can only recognise one touch at a time, so they are not suitable for multi-touch applications.

Capacitive.

- Under the glass of a capacitive screen, there is a layer of capacitive material.
- When a user touches the screen, a small amount of charge flows away from their finger because humans are conductive.
- The change in electrical charge is measured precisely: the closer the finger is to the charged areas, the more current flows away.
- This allows the computer to calculate the precise location at which the screen was touched.
- Capacitive touch screens are often used in smartphones.

BIOMETRIC SCANNER

- Biometric scanners work by measuring part of the unique physical characteristics of a user, often multiple times, until a suitable average result is produced.
- This average result is then stored and future samples are compared against it in order to check whether the person being scanned is the authorised one.

Fingerprint recognition: These scanners read the patterns of arches, loops and whorls in a human fingerprint

Facial recognition: These scanners identify the structure of a human face in order to identify an individual.

Voice recognition: These scanners require a microphone to capture the sound and a voice recognition software compares a voice print with a stored original and checks for a match.

Iris recognition: Like fingerprints, the human iris has a unique pattern, though an iris scan is approximately 120 times more detailed than a fingerprint. This means that iris recognition is 120 times more reliable.

Advantages and Disadvantages of using biometric Scanners.

ADVANTAGES

- Easier for users than passwords because biometric data cannot be forgotten
- More difficult to trick or confuse biometric scanners than some other methods
- Can speed up transactions
- Are not restricted by language barriers

DISADVANTAGES

- Cause privacy and security concerns because detailed personal information is stored electronically
- Possible to trick or confuse them
- Expensive to make and buy
- Make some people feel uncomfortable

CARD READERS

- Card readers are used to read data stored on a card that is carried by a user.
- The data can be used to unlock doors, access secure areas, make payments or track people, parcels and even pet animals.
- Cards can carry data using three methods. Each method needs a specialist type of reader

Magnetic strip: This is the least secure method as the data on the magnetic strip can easily be stolen by criminals who put the card through a card reader without the card owner's knowledge

Programmable chip: Data on a programmable chip is only readable when a correct PIN is entered into the reader.

Radio Frequency Identification (RFID) and Near Field Communication (NFC):

- RFID is a short-range wireless communication method, and Near Field Communication (NFC) is a branch of RFID.
- RFID tags are cheap and small and they can be included in a variety of objects such as cards, key fobs and smartphones.
- They contain a unique identification number (ID) that is linked to records in a database.



Sensors.

- Sensors are used to input data about the physical environment
- They can automatically input data without the need for human action.
- Their output is then processed by a computer.
- Sensors are one of the main

features of smart devices such as smartphones and smart environments such as smart homes



- The use of sensors has many benefits, such as:
 - they can be placed in remote or dangerous places
 - they can monitor continually
 - they remove the possibility of human error
 - they can sense things that people cannot (such as small changes in pressure or gas levels)
 - the data is easily (sometimes automatically) converted to a digital form.
- There are many different types of sensor to allow for the monitoring of a range of environmental factors. Some of these factors are:
 - light
 - moisture and humidity
 - Temperature
 - motion and movement
 - pressure.

OUTPUT.

Output peripherals are connected to a computer and output the results of the

computer's processing in various forms, including:

- electronic display
- printed text
- video
- audio
- tactile (touch) forms.



Output Devices of Computer



Monitor.

- Monitors allow users to see the output from the computer on an electronic display.
- Features of monitors include:
 - screen size, which is measured diagonally
 - resolution, which is measured in pixels
 - energy efficiency.
 - Some monitors also use touch screen technology.



PRINTER

There are three types of standard printer: dot matrix, inkjet and laser

Dot matrix:

- These printers are sometimes called "impact" printers.
- This is because the print is made by hitting or "impacting" the paper through a ribbon of ink, and this process is very noisy.
- The paper often has a number of carbon copy layers so it can make several copies of the same document as it prints it.





Inkjet printers:

- These printers use cartridges containing different coloured ink. Black ink is held in a one cartridge, while the colour cartridge is often split into three colours: cyan, magenta and yellow.
- Tiny dots of these inks are sprayed onto the paper to form images or text.

Laser printers:

- These printers contain a rotating cylinder or drum that holds an electrical charge.
- A laser is used to discharge certain points on the drum and 'draw' an image.
- Electrically charged toner is attracted to those points on the drum and is heated onto paper that is passed over the drum

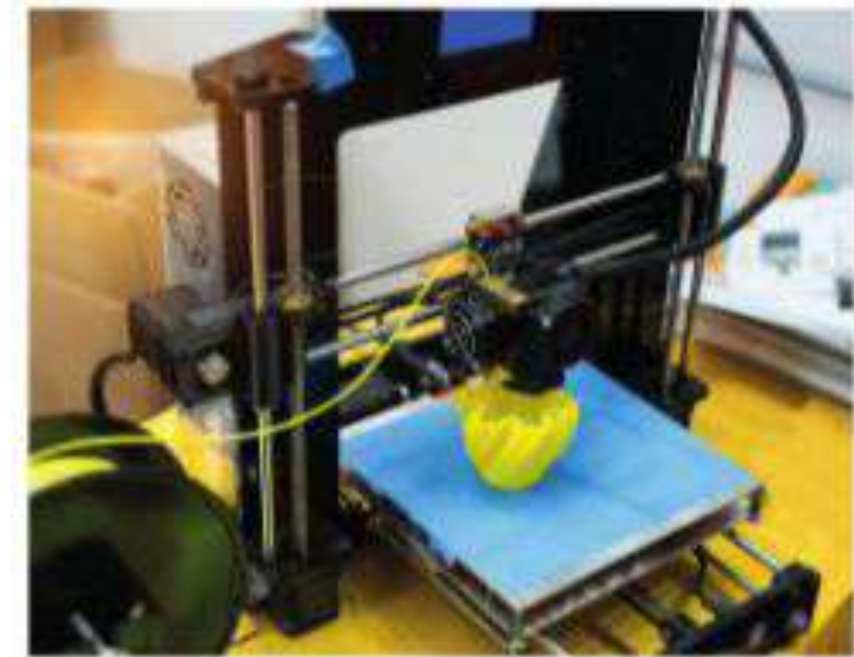


Comparing printer types

	Speed	Printer cost	Cost per copy	Colour printing	Quality
Dot matrix	Slowest	Cheap	Low	Rare	Low
Inkjet	Fast	Medium	High	Yes	High
Laser	Fastest	Expensive	Medium	Yes	Highest

3D PRINTER

- Three-dimensional (3D) printers create real-life versions of digital models
- These printers work by adding layers of heated material from the bottom of the model all the way to the top.
- 3D-printed models take a long time to create.



Plotter



- Plotters are used by a variety of professionals, including product designers, architects, engineers and cartographers.
- The plotter draws high-quality images created in Computer Aided Design (CAD) software onto large sheets of paper.
- It does this by moving a pen across the paper to draw the lines, and the pen can be raised away from and lowered onto the sheet of paper.
- Some plotters can also roll the paper backwards and forwards.

Data projectors.

- Data projectors are used for home entertainment, such as watching television or playing games, and for giving presentations in classrooms and business meetings.
- They have the following features
 - Light bulbs: These bulbs can have various brightness levels and are expensive to replace.
 - Resolution: Like monitors, they use different quality screens
 - Zoom functionality: The projected image can be made larger by enlarging the image using either optical zoom or digital zoom.
 - Portability: Some smaller and lighter projectors are available, and these are easier to move around.



Speakers.



- Speakers allow a computer to output sound.
- Speakers often come in pairs to provide stereo sound.
- Multiple speakers are commonly used to provide surround sound in home entertainment systems.
- These speakers need a special amplifier that can deliver different levels of sound to each speaker to provide spatial awareness.

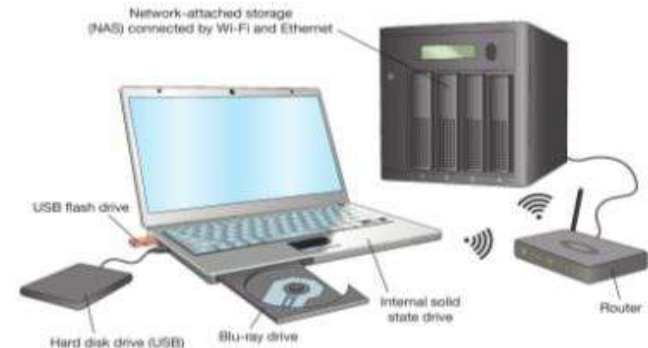
Control devices.

- Control devices are also known as actuators, and they are components of a system that make something happen in the real physical world.
- These devices are often mechanical.
- Examples of actuators include:
 - valves
 - pistons
 - heaters
 - coolers
 - motors.



STORAGE (Secondary storage)

- Secondary storage is often referred to as 'storage'. However, it should not be confused with primary storage or main memory, which are terms sometimes used to refer to Random Access Memory (RAM) or Read Only Memory (ROM)
- is non-volatile, which means it will not lose data when the system's power is turned off.
- Secondary storage is used for:
 - storing a document for future use
 - storing an application, ready to be used when the user opens the application
 - virtual memory



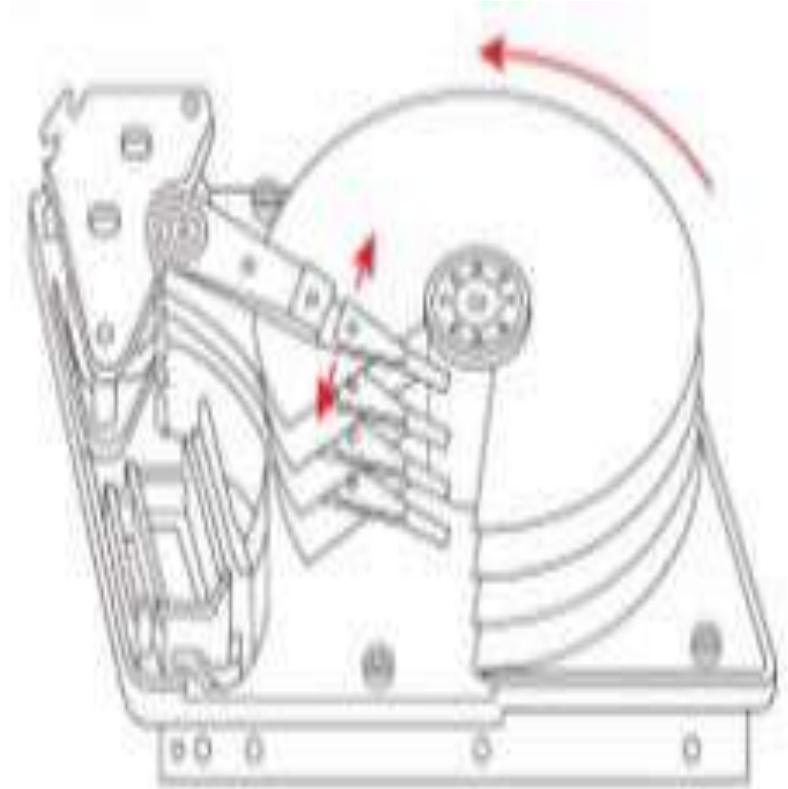
Storage devices and storage media.

- Storage devices are used to store data or software that is used in a computer system.
- Storage devices can be either internal or external. Internal drives connect directly to the computer's motherboard. External devices are connected to the computer's motherboard through the different ports and adapters on the computer's exterior
- the difference between a storage device and storage media is that a storage device contains the storage media and accesses the data stored on it. For example: flash memory is the media used to store data, whereas a USB drive is used to contain the flash media so that it can be read by the computer after it is plugged into a USB port

Hard disk drive

- Hard disks are made up of many concentric platters. These platters make up a cylinder that spins on a central spindle.
- A read/write head moves on an arm across tracks on the platter.
- The amount of time that it takes the read/ write head to access data on the tracks is determined by how fast the cylinder of platters spins and how fast the read/write head is moved across the tracks.
- When reading, a read/write head changes the magnetic field into electrical current.

- When writing, it transforms electrical current into a magnetic field.
- Typical spin speeds are 5400 revolutions per minute (rpm) or 7200 rpm.
- The average seek time for a read/write head is 4-15 milliseconds (ms)



Optical media.

- Types of optical media include CDs, DVDs and Blu-ray disks.

Compact disk (CD)

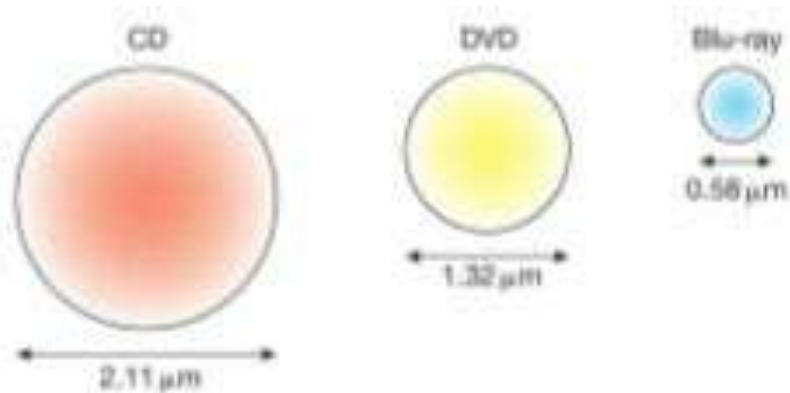
- CDs can store up to 700 MB of data.
- The data is written to the disk using a laser, which writes data to a plastic layer beneath layers of aluminium and acrylic.

Digital versatile disk (DVD)

- DVDs look very much like CDs.
- Data is written to DVDs using a shorter wavelength of red laser light, which allows DVDs to store more data.
- They can store 4.7 GB on a single-sided, single-layer disk, Double-sided, dual-layer disks can store as much as 18 GB.

Blue-ray disks

- Blu-ray disks are similar to CDs and DVDs, but use a scratch-protection coating.
- Violet laser light is used to store data at greater density than the red laser light used in DVDs.
- They can store 25 GB on a single-sided disk, and double-sided disks can store 50 GB.
- All optical media is recordable and is known as CD-R, DVD-R or BD-R. Some CDs and DVDs can be rewritten (CD-RW, DVD-RW, and some Blu-ray media Can be written then erased and rewritten (BD-RE).



Flash media

- Flash media are more energy-efficient than hard disks as they do not have moving parts.
- For the same reason, they are also less likely to fail when they are moved around. This makes them suitable for use in portable devices



Magnetic tape.

Magnetic tape was originally designed to record sound, but it is now also used to store data.



Comparing types of storage media

	HARD DISK	FLASH MEDIA	OPTICAL MEDIA	MAGNETIC TAPE
TRANSFER SPEED	Fast	Fastest	Slow	Slowest
MAXIMUM CAPACITY	Varies, up to 128 GB	Varies, up to many TB	CD 700 MB DVD 18 GB Blu-ray 50 GB	Varies, up to 185 TB
COST PER GB	High	Very high	Medium	Low
USE	Servers, personal computers, backups	Laptops, mobile devices	Multimedia (music, games and films), file backups	Whole system backups and archives
PORTABILITY	Not suitable	Yes	Not suitable	Not suitable

Storage capacity.

- capacity is the amount of data that can be stored on media
- bit (binary digit) is a single unit of information with a value of either 0 or 1
- eight bits equals one byte

Units of Computer Memory Measurements

1 Bit = Binary Digit

8 Bits = 1 Byte

1024 Bytes = 1 KB (Kilo Byte)

1024 KB = 1 MB (Mega Byte)

1024 MB = 1 GB (Giga Byte)

1024 GB = 1 TB (Terra Byte)

1024 TB = 1 PB (Peta Byte)

1024 PB = 1 EB (Exa Byte)

1024 EB = 1 ZB (Zetta Byte)

1024 ZB = 1 YB (Yotta Byte)

1024 YB = 1 (Bronto Byte)

1024 Brontobyte = 1 (Geop Byte)

Geop Byte is The Highest Memory

Homework.

1. State one benefit of using barcodes in a supermarket.
2. State one reason why small, built-in microphones are used in smartphones and laptops.
3. Explain why capacitive touch screens do not work when you are wearing woolen gloves.
4. State three computer systems that do not use monitors
5. State two benefits of using laser printers rather than inkjet printers.
6. State two types of business that use plotters
7. Explain why some 3D printers use biodegradable materials to create models
8. List two sensors used in smartphones. For each one, describe one way in which it is used by the operating system or an app.
9. Discuss the advantages and disadvantages of using sensors in smart
10. Explain one reason why Blu-ray disks are used to store modern console games

END

Thanks for reading